

network_sweep: TODO4v2 vs. 7 BEELINE competitors

t1=1. One column per topology family (source: summary_metrics_table_network_sweep_t1_1.csv). e17_pos100_density excluded. 9-edge topology reused as the ‘all positive’ negative fraction column.

0.00.20.40.60.81.0

score

Signed directed — AUPRC

Network density		Fraction of negative edges						Avg.	Rank
		5 edges	9 edges	13 edges	Yes negative	Balanced	All positive		
TODO4v2		.74	.62	.52	.47	.61	.62	0.59	#1
	Pearson	.37	.56	.51	.44	.46	.56	0.49	#2
	PPCOR	.35	.52	.51	.45	.44	.52	0.47	#3
	PIDC	.32	.56	.53	.07	.28	.56	0.39	#4
	GRNBoost2	.30	.58	.52	.07	.24	.58	0.38	#5
	scSGL	.20	.31	.33	.28	.47	.31	0.32	#6
	SCODE	.17	.33	.46	.07	.13	.33	0.25	#7
	GENIE3	.12	.29	.47	.07	.19	.29	0.24	#8

Unsigned directed — AUPRC

		Network density			Fraction of negative edges			Avg.	Rank
		5 edges	9 edges	13 edges	Yes negative	Balanced	All positive		
TODO4v2		.74	.63	.60	.49	.62	.63	0.62	#1
	Pearson	.37	.57	.57	.47	.48	.57	0.51	#2
	PPCOR	.36	.56	.56	.49	.47	.56	0.50	#3
	PIDC	.32	.56	.53	.40	.48	.56	0.47	#4
	GRNBoost2	.30	.58	.52	.32	.44	.58	0.46	#5
	scSGL	.20	.37	.48	.35	.49	.37	0.38	#6
	SCODE	.17	.33	.46	.36	.33	.33	0.33	#7
	GENIE3	.12	.29	.47	.26	.40	.29	0.30	#8

Unsigned undirected — AUPRC

		Network density						Fraction of negative edges		Avg.	Rank
		5 edges	9 edges	13 edges	Yes negative	Balanced	All positive				
TODO4v2		.81	.78	.77	.71	.88	.78			0.79	#1
	Pearson	.68	.80	.74	.69	.82	.80			0.76	#2
	PPCOR	.67	.79	.73	.70	.82	.79			0.75	#3
	GRNBoost2	.67	.80	.67	.60	.80	.80			0.72	#4
	PIDC	.59	.81	.72	.58	.80	.81			0.72	#5
	scSGL	.38	.55	.65	.56	.83	.55			0.59	#6
	SCODE	.41	.62	.71	.53	.64	.62			0.59	#7
	GENIE3	.31	.58	.74	.45	.63	.58			0.55	#8

Signed directed — F1 (top-k)

		Network density						Fraction of negative edges		Avg.	Rank
		5 edges	9 edges	13 edges	Yes negative	Balanced	All positive				
TODO4v2		.61	.54	.47	.46	.59	.54		0.53	#1	
	Pearson	.34	.51	.49	.44	.47	.51		0.46	#2	
	PPCOR	.35	.46	.48	.45	.46	.46		0.44	#3	
	PIDC	.28	.56	.51	.00	.30	.56		0.37	#4	
	GRNBoost2	.27	.55	.48	.00	.23	.55		0.35	#5	
	scSGL	.19	.32	.31	.28	.50	.32		0.32	#6	
	SCODE	.11	.31	.44	.00	.06	.31		0.20	#7	
	GENIE3	.04	.24	.45	.00	.11	.24		0.18	#8	

Unsigned directed — F1 (top-k)

		Network density			Fraction of negative edges			Avg.	Rank
		5 edges	9 edges	13 edges	Yes negative	Balanced	All positive		
TODO4v2		.61	.54	.51	.47	.59	.54	0.54	#1
	Pearson	.34	.51	.53	.47	.47	.51	0.47	#2
	PIDC	.28	.56	.51	.40	.50	.56	0.47	#3
	PPCOR	.35	.46	.52	.49	.46	.46	0.46	#4
	GRNBoost2	.27	.55	.48	.28	.41	.55	0.42	#5
	scSGL	.19	.35	.46	.33	.50	.35	0.36	#6
	SCODE	.11	.31	.44	.30	.29	.31	0.29	#7
	GENIE3	.04	.24	.45	.24	.35	.24	0.26	#8

Unsigned undirected — F1 (top-k)

		Network density						Fraction of negative edges		Avg.	Rank
		5 edges	9 edges	13 edges	Yes negative	Balanced	All positive				
TODO4v2		.76	.66	.68	.64	.80	.66			0.70	#1
	Pearson	.73	.68	.66	.66	.75	.68			0.69	#2
	GRNBoost2	.69	.71	.63	.57	.72	.71			0.67	#3
	PIDC	.55	.74	.66	.57	.75	.74			0.67	#4
	PPCOR	.71	.57	.65	.64	.70	.57			0.64	#5
	scSGL	.35	.52	.63	.56	.72	.52			0.55	#6
	SCODE	.35	.58	.65	.48	.57	.58			0.54	#7
	GENIE3	.24	.54	.67	.42	.60	.54			0.50	#8